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Digital twin-based structural health monitoring and measurements of dynamic characteristics in balanced cantilever bridge

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ABSTRACT

This study developed a digital twin (DT) and structural health monitoring (SHM) system for a balanced cantilever bridge, utilizing advanced measurement techniques to enhance accuracy. Vibration and dynamic strain measurements were obtained using accelerometers and piezo-resistive strain gauges, capturing low-magnitude dynamic strains during operational vibrations. 3D-LiDAR scanning and Ultrasonic Pulse Velocity (UPV) tests captured the bridge's as-is geometry and modulus of elasticity. The resulting detailed 3D point cloud model revealed the structure's true state and highlighted discrepancies between the as-designed and as-built conditions. Dynamic properties, including modal frequencies and shapes, were extracted from the strain and acceleration measurements, providing critical insights into the bridge's structural behavior. The neutral axis depth, indicating stress distribution and potential damage, was accurately determined. Good agreement between vibration measurement data and the as-is model results validated the reliability of the digital twin model. Dynamic strain patterns and neutral axis parameters showed strong correlation with model predictions, serving as sensitive indicators of local damage. The baseline digital twin model and measurement results establish a foundation for future bridge inspections and investigations. This study demonstrates the effectiveness of combining digital twin technology with field measurements for real-time monitoring and predictive maintenance, ensuring the sustainability and safety of the bridge infrastructure, thereby enhancing its overall resilience to operational and environmental stressors.

1. Introduction

Ensuring the safety, longevity, and functionality of bridges, particularly as they age within crucial transportation networks, is a critical aspect of infrastructure management focused on enhancing their overall resilience to various stressors [1]. Structural health monitoring (SHM) plays a key role in achieving this by continuously assessing structural integrity and performance, enabling early detection of issues such as cracks, corrosion, and unusual vibrations that could lead to catastrophic failures [2–4]. A proactive SHM approach facilitates timely maintenance and repairs, mitigating risks associated with accidents, downtime, and excessive costs. However, traditional inspection methods—such as visual surveys and single-point measurements—often fail to capture the high-resolution, real-time data required for effective long-term monitoring [5,6]. This limitation is particularly significant for modern bridge structures with complex geometries and dynamic behaviors, where localized failures may not be immediately apparent.

Recent advancements in SHM have increasingly leveraged digital twin (DT) technology, which integrates real-time data with numerical simulations to create a dynamic, continuously updated representation of a structure [7,8]. By providing near real-time insights into structural behavior, DTs enhance predictive maintenance and decision-making, ensuring that bridges remain resilient under increasing traffic demands and challenging environmental conditions [9]. However, implementing DT models effectively requires tailoring them to different bridge types, each of which presents unique structural complexities and monitoring challenges. Various structural systems—such as balanced cantilever, cable-stayed, and suspension bridge exhibit distinct geometric configu-

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rations and load distributions, influencing sensor placement, data acquisition strategies, and model accuracy. While DT applications have shown promise for certain bridge types, such as cable-stayed bridges [10], significant gaps remain in their application to more geometrically complex systems.

Balanced cantilever bridges are widely used for spanning large distances due to their efficient use of materials and construction techniques. They are often deployed in environments where conventional bridge construction methods may not be feasible. Despite these advantages, they pose significant monitoring challenges due to their segmental construction, variations in cross-sectional geometry, and susceptibility to dynamic loads [11,12]. Their structural behavior under different loading and environmental conditions is highly complex, making real-time data integration essential for reliable SHM. However, traditional inspection approaches often lack the capacity to capture the time-sensitive, high-resolution data required for effective long-term monitoring. While digital twins offer a promising solution, fully capitalizing on a DT model that accurately integrates real-time geometric changes and evolving material properties remains an open challenge.

Addressing this research gap requires a robust approach that integrates advanced measurement techniques with high-fidelity numerical modeling to improve the accuracy and reliability of balanced cantilever bridge monitoring. The challenge lies not only in real-time data integration but also in refining DT frameworks to accommodate the unique geometric and dynamic complexities of these structures. Successfully overcoming these challenges will enhance predictive maintenance, improve structural assessments, and ensure the long-term safety of balanced cantilever bridges.

This study aims to address these challenges by developing a DT-based SHM system specifically for balanced cantilever bridges, leveraging advanced measurement techniques and digital modeling to enhance accuracy and reliability, thereby fostering a more resilient and sustainably managed infrastructure asset [13]. A comprehensive digital twin model requires precise measurements of the bridge's dynamic behavior and structural responses. To achieve this, the study utilizes vibration and dynamic strain measurements obtained through accelerometers and piezoresistive strain gauges [14,15]. Piezo-resistive strain gauges, which can capture low-magnitude dynamic strains during operational vibrations, provide critical data for model calibration. By integrating these detailed measurements into the digital twin model, this study seeks to develop a more accurate representation of the bridge's real-time behavior, thereby enabling a more effective SHM strategy [16,17].

The incorporation of 3D-LiDAR scanning and Ultrasonic Pulse Velocity (UPV) tests marks a significant advancement in capturing the as-is geometry and material properties of the bridge [18,19]. The 3D point cloud model generated from the LiDAR scan offers a comprehensive and detailed visualization of the bridge, surpassing the capabilities of conventional visual inspections, especially in terms of time efficiency, accessibility, and detail [20]. These advanced techniques ensure that the digital twin model reflects the true state of the structure, highlighting discrepancies between as-designed and as-built conditions, and emphasizing the importance of updating structural models throughout the service life of the infrastructure [21].

Dynamic properties derived from strain and acceleration measurements provide crucial insights into the structural behavior under operational conditions [22,23]. Strain gauges capture localized deformation, while accelerometers provide global response characteristics, such as modal frequencies and shapes. The neutral axis depth is a critical parameter, indicating the stress distribution and potential damage within the structure [24,25]. Discrepancies between the as-is measurements and as-designed drawings highlight the importance of accurate, real-time data in maintaining structural integrity. These discrepancies may arise from material degradation, construction deviations, or unforeseen loading conditions, necessitating regular updates to the digital twin model for reliable SHM and predictive maintenance.

Furthermore, the validation of the digital twin model through comparison with field measurement data is a critical step in ensuring its accuracy [26,27]. This study conducts extensive vibration measurements to extract modal properties, such as natural frequencies and bending mode shapes, which are then compared with the numerical model results [28,29]. Good agreement between the measurement data and the as-is model results underscores the reliability of the digital twin approach. The dynamic strain patterns and neutral axis parameters, sensitive indicators of local damage, are also analyzed and found to correlate well with the model predictions [30,31]. These findings support the use of digital twin models and field measurements as baseline data for future bridge inspections and investigations, while also identifying areas for further refinement, particularly at span join locations [32,33].

2. Digital twin-based structural health monitoring

Fig. 1 presents the framework for a combined Digital Twin (DT) and structural health monitoring system designed for the maintenance of a cantilever bridge. The left side outlines the DT model, which is separated into two segments: the As-designed and the As-is models. The As-designed model is based on the bridge's initial design details, including a 3D Finite Element Model (3D-FEM) and its initial dynamic properties. On the other hand, the As-is model updates the FEM to reflect real-time conditions using data such as the concrete's modulus of elasticity obtained through UPV testing and the bridge's geometry captured by 3D-LiDAR scans. This results in updated dynamic properties of the bridge, which are essential for accurate assessment and ongoing monitoring.

Concerning the structural health monitoring component (see the right side of Fig. 1), it uses dynamic measurements like acceleration and dynamic strain, obtained through accelerometers and piezoresistive strain gauges, respectively. The data collected from these devices are then processed for data acquisition purposes. These experimental measurements validate and update the dynamic properties in the digital twin model. The footnote clarifies that the dynamic properties include frequency, mode shapes, normalized modal strain (NMS), and neutral axis depth. The integration of the digital twin model with experimental data from structural health monitoring allows for a comprehensive assessment of the bridge's condition and serves as a baseline for future comparisons, ensuring an informed approach to maintenance and management.

While the individual components of this process, such as numerical modeling and structural measurements, are well-established techniques, their integration within a digital twin framework —which inherently involves an interplay between the virtual model and the physical asset to build resilience [34]—enables a more powerful, data-driven approach to continuous model updating, performance tracking, and predictive maintenance aligning with advancements in DT modeling that emphasize comprehensive data integration [35,36]. The DT framework is particularly valuable when construction quality is uncertain or when deviations from design specifications may exist, as it provides a dynamic and traceable baseline throughout the service life of the infrastructure. Transitioning from the theoretical framework to practical application, the integration of the DT model with real-time data from structural health monitoring, as described in this section, sets the stage for its deployment on actual infrastructure. The subsequent section focuses on a case study involving the targeted balanced cantilever bridge in Thailand, highlighting the real-world utility of the technology.

3. Structural health monitoring

3.1. Targeted balanced cantilever bridge

In this study, a 10-year-old bridge employing the balanced cantilever system and crossing a canal in Nonthaburi Province, Thailand, was selected. With a total span of 210 m, three pier-to-pier spans (gridlines E1-E2, E2-E3 and E3-E4) and the closure segments were placed at the

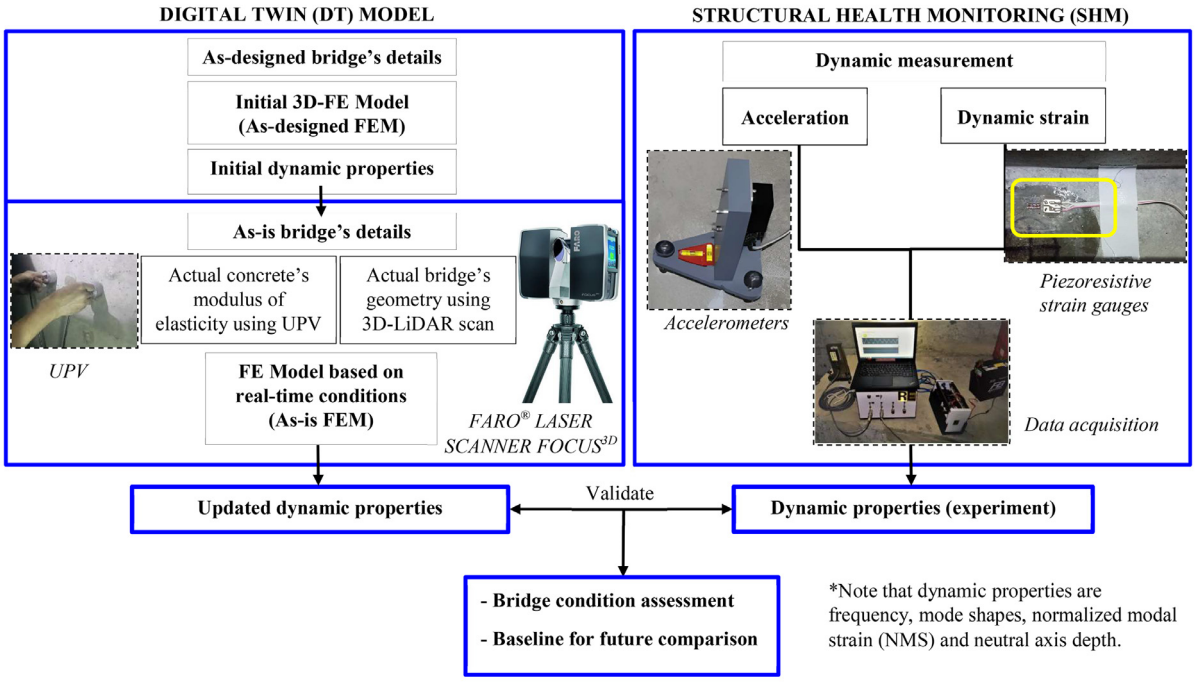


Fig. 1. Framework combining DT technology with SHM for the targeted balanced cantilever bridge.

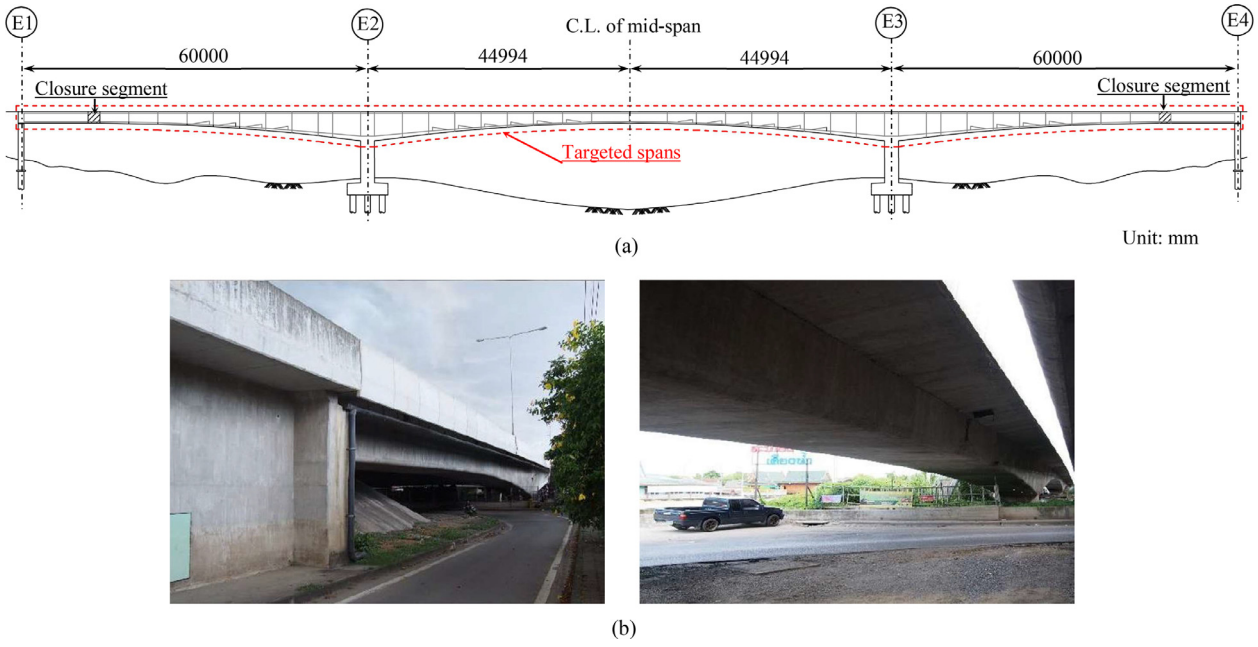


Fig. 2. The targeted bridge; (a) As-designed drawing and (b) Actual bridge.

both-end spans (gridlines E1-E2 and E3-E4), as depicted in Fig. 2. These precast segmental concrete bridges involve multiple concrete segments with varying depths, connected by post-tensioning and wet joints using non-shrink grout or high-strength mortar. While these precast segments facilitate rapid construction, cost-efficiency, and excellent quality control, the joints cause significant discontinuities that can impact the overall structural behavior of segmental bridges. Fig. 3 shows the typical cross-section of the segmental prestressed box girders, where the depths "H" ranges from 2 m to 5 m, the width "B" ranges from 3.7 m to 4.3 m, and the thickness "HB" of the bottom slab ranges from 0.25 m to 0.8 m. Following the placement of the segmental box girders, an adjacent 1.6-meter-wide cantilever slab for the walkway was constructed using cast-in-place concrete. This as-designed drawing will then be used to construct the analytical model in order to obtain dynamic properties of the targeted bridge.

3.2. 3D-LiDAR scan

In the DT model, the updating geometry is necessary to assess the real-time or as-is conditions of the structure, especially discontinuities at the joint connections. Therefore, the targeted bridge was scanned using 3D-LiDAR technique (see FARO® LASER SCANNER FOCUS^{3D} [37]

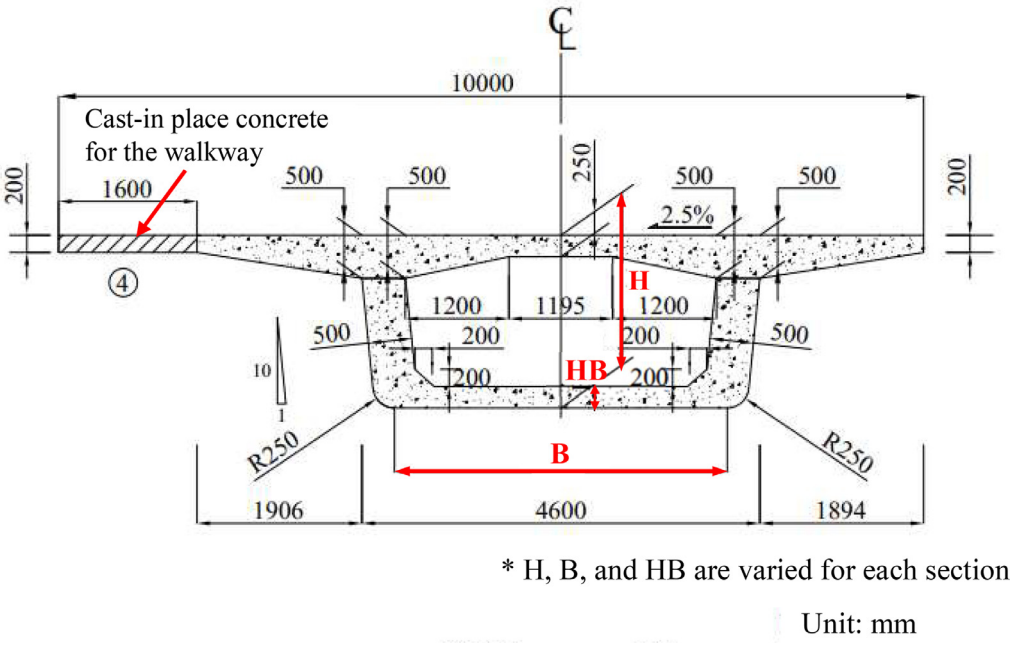


Fig. 3. Typical section of the segmental prestressed box girder.

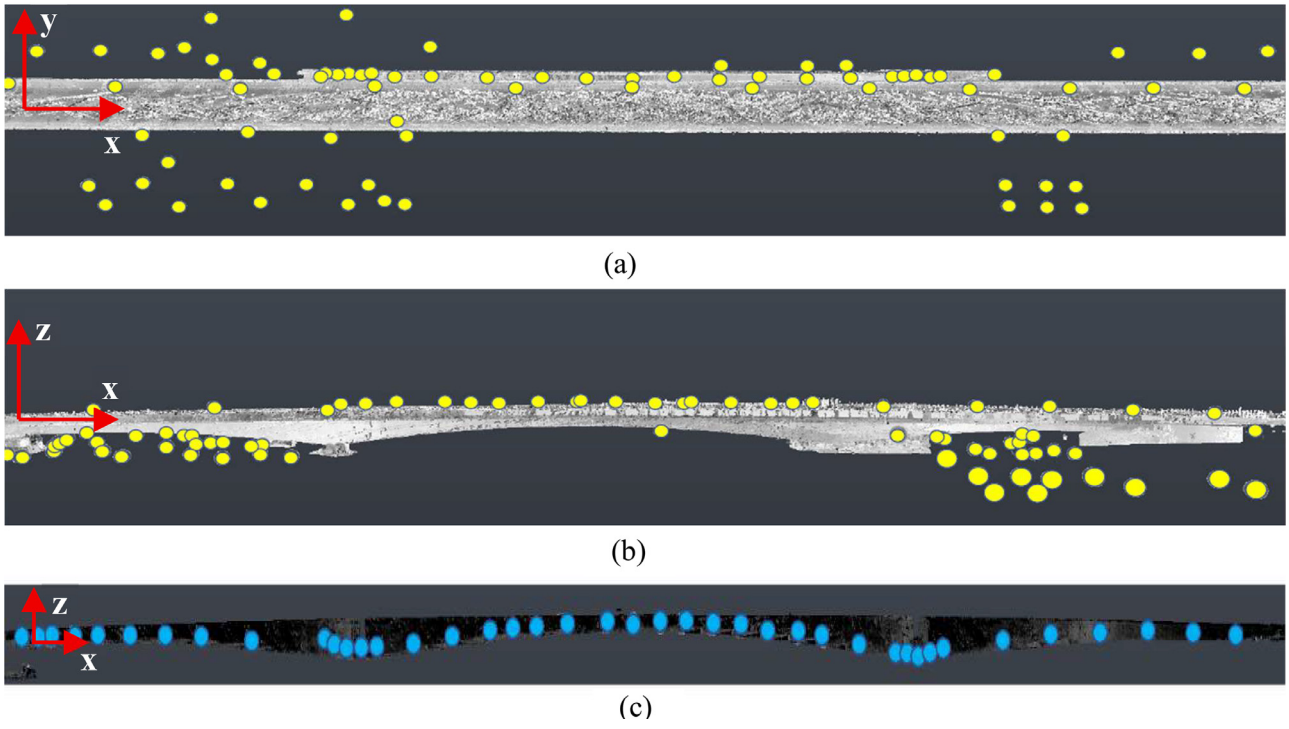
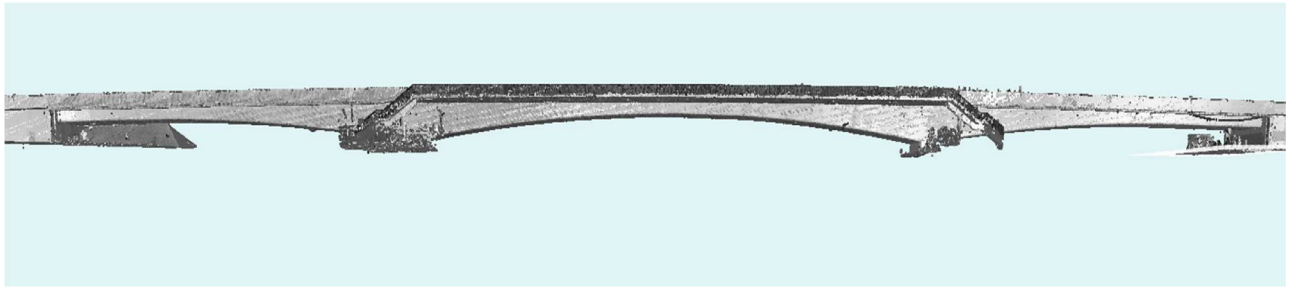
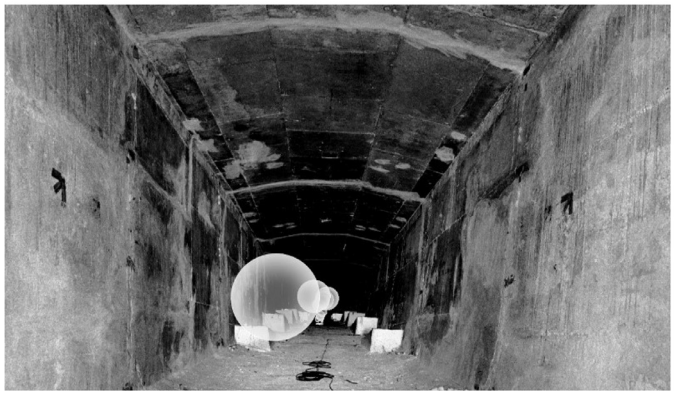


Fig. 4. Locations of 3D-LiDAR scanning stations (indicated by spheres) relative to the targeted bridge: (a) overall top view of station distribution; (b) front view showing stations primarily for scanning outer box girder surfaces; and (c) front view showing stations primarily for scanning inner box girder surfaces.

in Fig. 1). The bridge was captured using a total of 170 scanning stations, with their spatial distribution illustrated in Fig. 4. The coordinate system adopted for referencing these scans and the resulting point cloud consists of the X-axis aligned longitudinally along the bridge (positive to the right as viewed in plans/elevations), the Y-axis as transverse, and the Z-axis as vertical (positive upwards). Fig. 4(a) provides a top view of all station locations, while Fig. 4(b) and (c) offer front views highlighting stations used for scanning the outer and inner surfaces of the box girders, respectively. The 3D-LiDAR scan had a measurement range from 0.6 to 76 m and a resolution of ±3 mm at a distance of 25 m. A total of 41 stations were used for the outer box girder scans (typically shown as yellow spheres in Fig. 4) and 129 stations for the inner box girder scans (typically shown as blue spheres in Fig. 4). Four reference targets were attached to the entrance of the box girder on the North side of the bridge to accurately align the point clouds from the inner and outer surface scans within this defined coordinate system. The process of collecting the bridge's geometric data was conducted during normal traffic flow on the bridge.



(a)



(b)



(c)

Fig. 5. Details of the bridge from a 3D-LiDAR scan; (a) overview of the bridge, (b) inner box girders, and (c) outer box girders.

After aligning the point clouds from each station onto the global point cloud data in Autodesk ReCap software, noise reduction, and outlier removal processes were applied to enhance data quality. Fig. 5 presents the resulting optimized point cloud of the bridge, where Fig. 5(a) shows an overall view of the captured structure. Fig. 5(b) displays a view from the outer box girder, and Fig. 5(c) provides a view from the inner box girder. The spherical objects visible in Figs. 4 and 5 represent the LiDAR scanner locations. Since each individual scan has a limited field of view, multiple scanning stations were required to capture the full geometry of the structure. These sphere visuals are automatically generated by the post-processing software to indicate scanner positions and assist in point cloud registration and alignment. Following point cloud optimization, the geometric details were modeled in Autodesk Revit and used as a baseline for structural evaluation. The finalized geometry was then imported into SAP2000 [38] to construct a 3D finite element model (FEM), enabling numerical simulation and analysis of the bridge's current structural behavior.

3.3. 3D-LiDAR extraction and comparison of as-designed and as-is bridge geometry

The registration and noise reduction of point clouds from a total of 170 scanning stations were performed using Autodesk Recap software to extract the geometric data. Subsequently, the extracted geometry data was evaluated and compared with the as-designed drawings, focusing on several key dimensions as illustrated in Fig. 6(a). The key dimensions critical to the bridge girders's bending moment capacity were designated as "A" for the top slab width, "B" for the box girder depth, "C" for the bottom slab width of the box girder, and "D" for the bottom slab depth of the box girder. These dimensions are crucial for determining the structural integrity and load-bearing capabilities of the bridge girders.

Through the comparison of section pairs, deviations from symmetry can indicate construction errors or post-construction deformations, affecting the bridge's performance. Therefore, the sections labeled F1'-F1' and F2'-F2' are supposed to be symmetrical counterparts to F1-F1 and F2-F2, respectively (Fig. 6(b)). In Fig. 6(c), the dotted lines represent the as-designed drawings which are the intended structural shapes according to the bridge's design specifications. Meanwhile, the solid lines, shown in blue, illustrate the as-is condition derived from 3D-LiDAR scan. Areas where the 3D-LiDAR scan diverges from the as-designed drawings are evident, potentially indicating construction defects, material degradation, or other deviations from the original design. Table 1 summarizes the percentage difference of the geometry obtained from as-designed and as-is drawings, showing an error varying between –0.02 % to +2.59 %.

The bridge profile, referring to the side view outline of the bridge, was extracted from the 3D-LiDAR scan to compare with that of the as-designed drawing. In Fig. 7, reference points positioned at the central apex of the top surfaces over the pier supports - where variance in displacement is markedly lower - are employed for comparison of the bridge profile between 3D-LiDAR and as-designed drawing. The notable discrepancies in the alignment of the bridge's top and bottom surfaces can be observed, especially at the midsection and both ends of the bridge. At the bridge's midpoint, the greatest variance in depth reached 5.86 %. Furthermore, the local thicknesses of the top and bottom deck flanges exhibited considerable variation, with differences amounting to 28.98 % and 15.76 %, respectively. Discrepancies in bridge deck depth and flange thickness may result from conditions experienced during construction or throughout the bridge's service life, prompting the detailed development of DT models that reflect the as-is condition. This approach places additional emphasis on the joint connections and the current alignment of the bridge to ensure more accurate modeling and analysis.

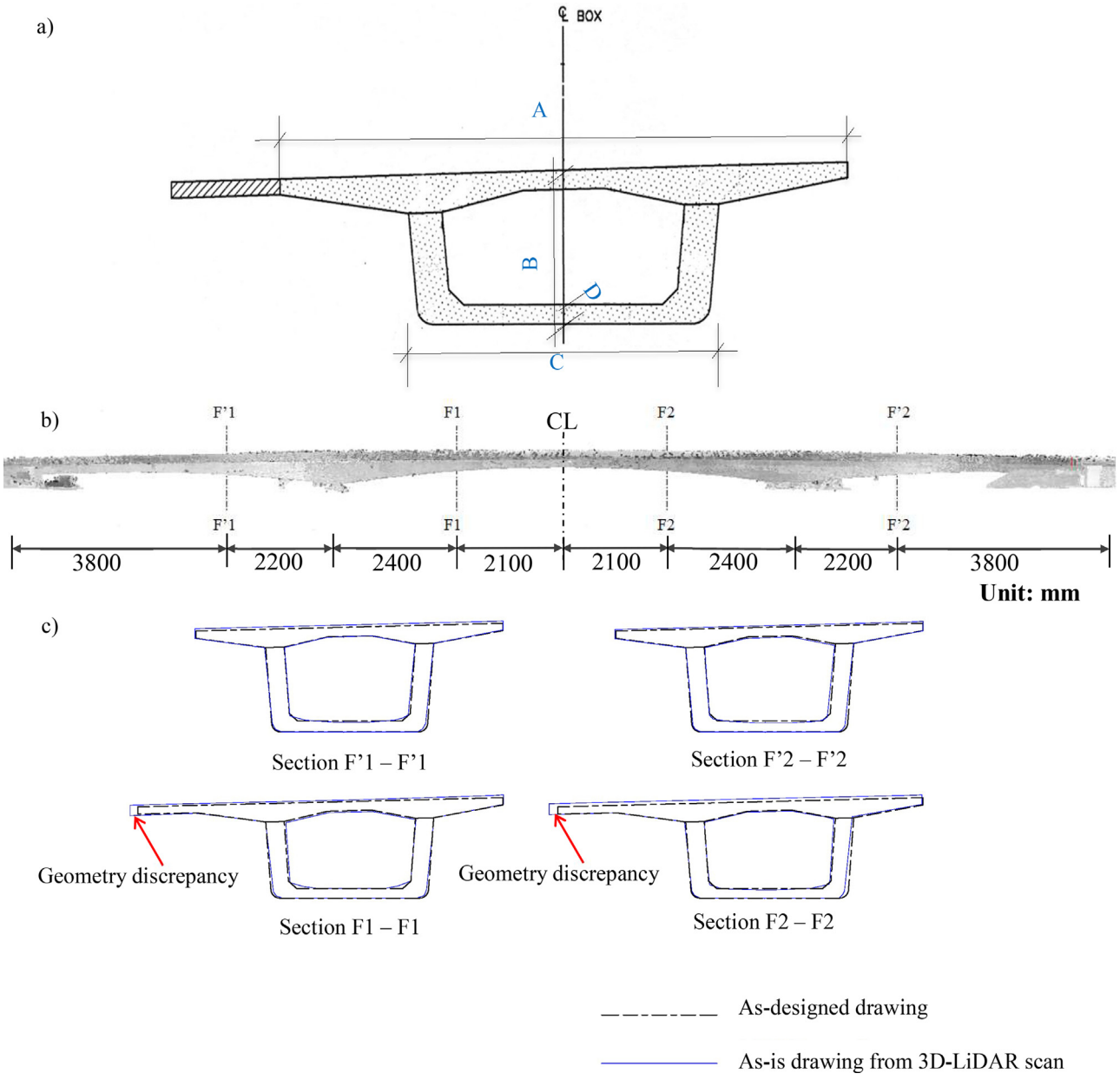


Fig. 6. Details of bridge geometry and discrepancies; (a) selected key dimensions at each section extracted from the 3D-LiDAR scan, (b) section profiles along the bridge, indicating measurement points, and (c) geometry discrepancies between as-designed and as-is drawings, highlighting deviations at specific sections.

3.4. Ultrasonic pulse velocity (UPV) test

In this study, the UPV technique (see Fig. 1) was employed to determine the current properties of the concrete, which will subsequently be incorporated into the 3D-FEM. This technique is a non-destructive testing method that evaluates the quality, integrity, and uniformity of concrete, masonry, and other construction materials. By measuring the time, it takes for ultrasonic pulses to travel through a material, UPV can determine its properties and identify defects, voids, or signs of deterioration without inflicting any damage. It should be noted that the targeted bridge, which has been in service for 10 years, may have material properties that deviate from the original design values over its service life [12,39]. Notably, the concrete's modulus of elasticity, a critical and time-dependent parameter, plays a significant role in determining the structural capacity and modal characteristics of the bridge.

To estimate the modulus of elasticity, the following empirical relationship was adopted, correlating ultrasonic pulse velocity (UPV) with the dynamic modulus of elasticity [40]:

$$E_d = \rho V^2 \frac{(1+\upsilon)(1-2\upsilon)}{(1-\upsilon)}$$

where E_d is the dynamic elastic modulus, ρ is the density of concrete, V is the ultrasonic pulse velocity, and ν is Poisson's ratio. In this study, a typical value of 0.2 was assumed for Poisson's ratio, as commonly accepted for normal concrete in the absence of precise experimental data. The influence of variation in Poisson's ratio on the estimated modulus was considered minimal for the intended application.

The concrete's surface along the bridge was tested using a Tico Proceq device equipped with 54 kHz transducers. The UPV tests using lubricant grease for surface attachment were conducted at 40 selected locations at 5-meter intervals, as illustrated in Fig. 8(a). At each of these locations, the UPV test was recorded at two webs inside the box girder. The modulus of elasticity, interpreted from the UPV test results, is displayed in Fig. 8(b) and ranged from 30,385 MPa to 45,433 MPa at various points along the bridge. This contrasts with the 29,436 MPa concrete's

Table 1
Comparison of geometry obtained from as-designed drawing and 3D-LiDAR scan.

Section	Key dimension	Geometry		Percentage difference (%)
		As-designed drawing (L_{design}, mm)	3D-LiDAR scan (L_{scan}, mm)	
F'1	A	8400.0	8427.9	0.33
	B	2861.0	2935.2	2.59
	C	4600.0	4609.1	0.20
	D	300.0	298.4	−0.53
F1	A	8400.0	8429.9	0.36
	B	2638.0	2679.3	1.57
	C	4600.0	4604.4	0.10
	D	265.0	266.3	0.49
F2	A	8400.0	8422.1	0.26
	B	2638.0	2691.5	2.03
	C	4600.0	4605.3	0.12
	D	265.0	267.9	1.09
F'2	A	8400.0	8420.8	0.25
	B	2861.0	2904.3	1.51
	C	4600.0	4599.0	−0.02
	D	300.0	297.5	−0.83

Note that the percentage difference can be calculated by $(\frac{L_{scan}-L_{design}}{L_{design}})\times 100$.

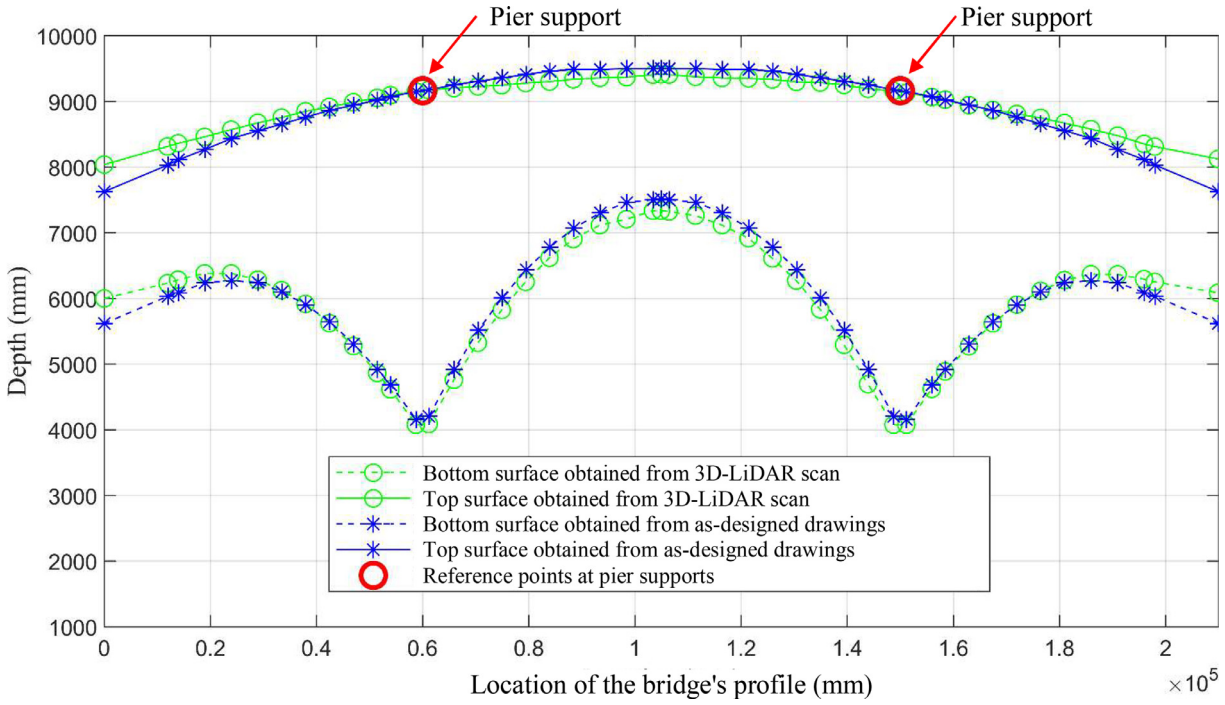


Fig. 7. Comparison of the bridge profiles obtained from 3D-LiDAR scan and as-designed drawing.

modulus of elasticity specified in the as-designed drawings. Therefore, the modulus of elasticity measured at various locations along the bridge was used to update the DT model to reflect the as-is condition.

3.5. 3D-FEM in the DT model of the bridge

To investigate the modal properties of the bridge, the 3D-FEM for both the as-designed and as-is conditions was simulated using modal analysis with SAP2000 [38]. The as-designed FEM was developed based on geometric and concrete properties specified during the design stage. In contrast, the as-is FEM incorporated geometric details captured from the 3D-LiDAR scan and the concrete's elastic modulus from UPV to reasonably depict the current condition. In the 3D-FEM, the segmental box girders and walkway were modeled using shell elements, while the mid-span piers were represented by solid elements, as depicted in Fig. 9(a). The shell elements are four-node area objects that reflect plate-bending behavior with assigned edge constraints, illustrated in Fig. 9(b). Meanwhile, the solid elements are defined as eight-node objects, each having six quadrilateral faces with a joint at each corner, as presented in Fig. 9(c). The model was set with a mesh size of 250 mm. Structural constraints were assigned to the pier-to-deck joint to ensure equal rotation and translation of connecting regions.

For both the as-designed and as-is conditions, roller systems were employed at both ends for support, and lateral springs were installed at the pier ends of the mid-span to represent the lateral and vertical stiffness of the piles. In the as-designed model, the modulus of elasticity of concrete for the segmental box girders and piers were 29,436 MPa and 27,534 MPa, respectively. Since the UPV test results showed that the modulus of elasticity for each box segment varied between 30,385 MPa to 45,433 MPa, the concrete's modulus of elasticity at each box segment was updated in the as-is model. Then, the models generated updated dynamic properties such as natural frequencies, mode shapes, and

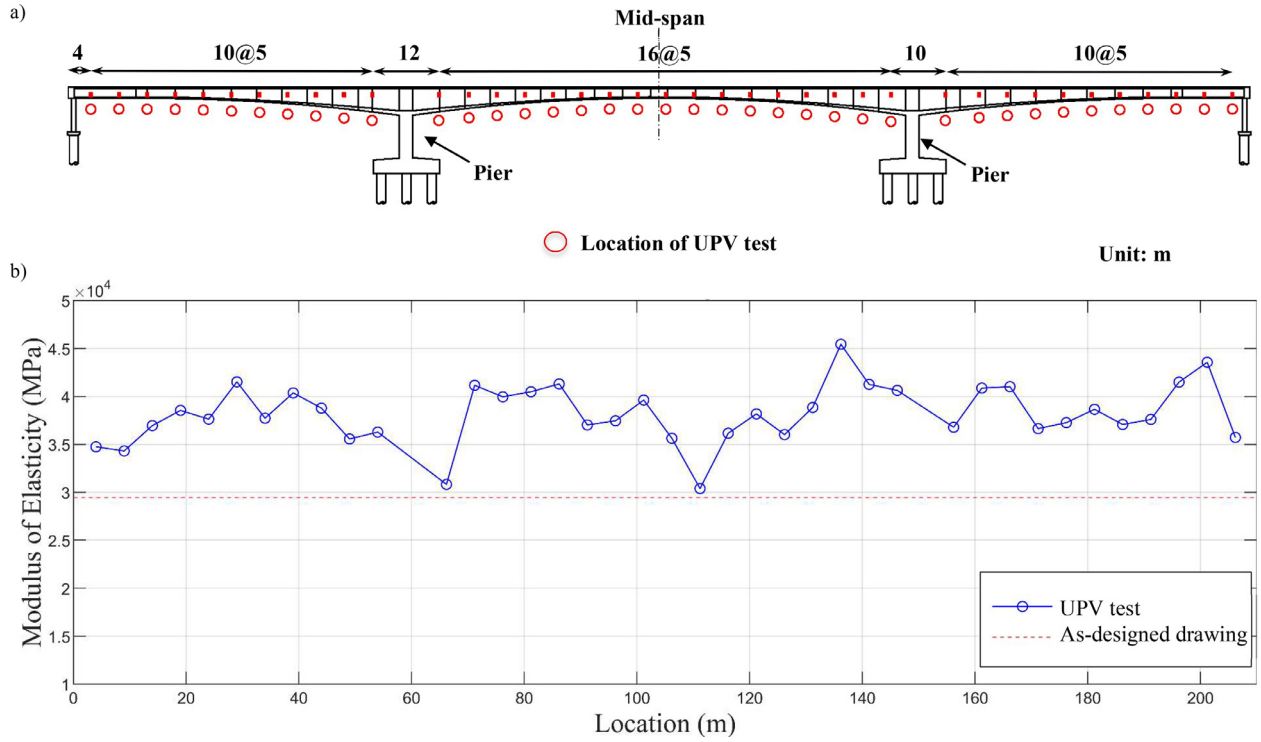


Fig. 8. Details of UPV test and concrete elasticity; (a) location of UPV test along the longitudinal direction and (b) elastic modulus of concrete interpreted from UPV test.

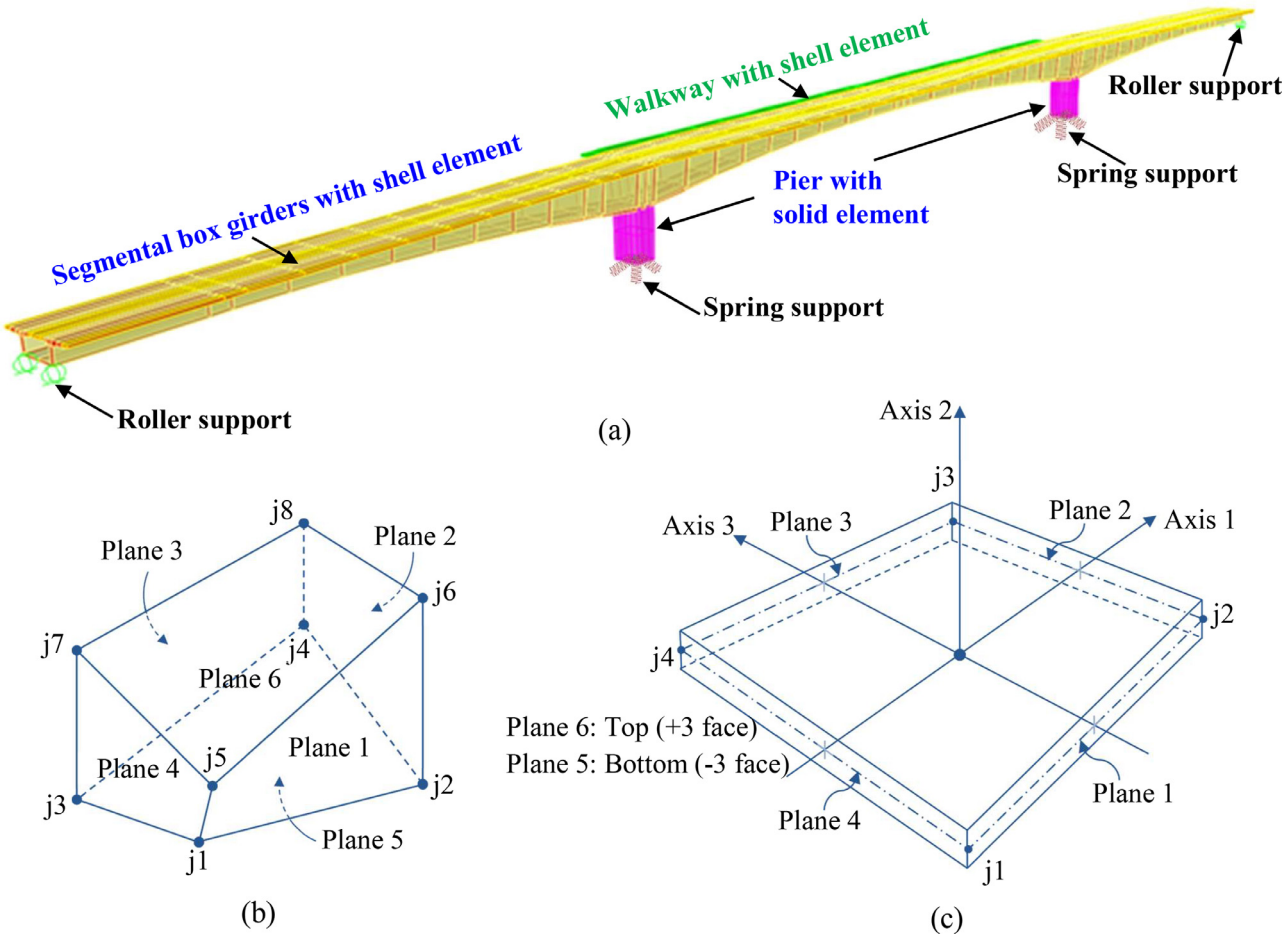


Fig. 9. Modal updating of the bridge using SAP2000; (a) 3D modelling, (b) Solid element, and (c) Shell element.

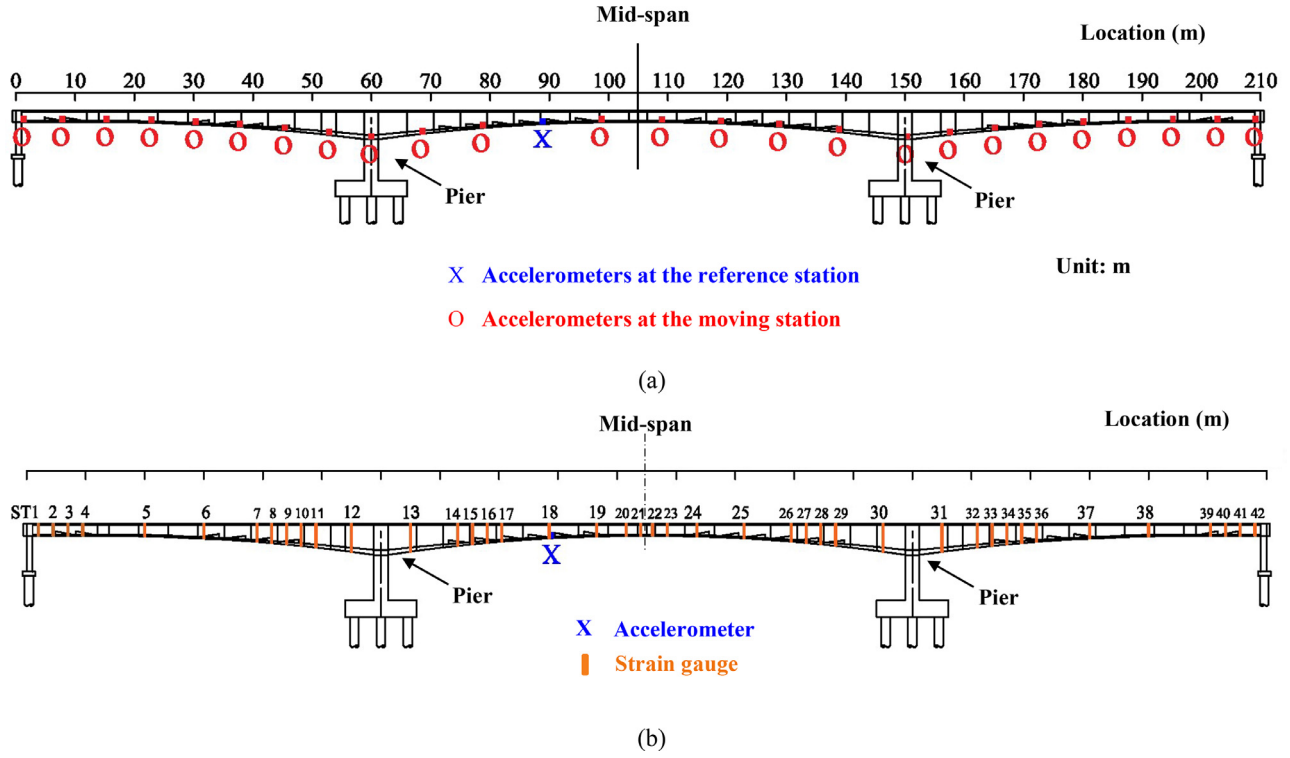


Fig. 10. Arrangement of accelerometers and strain gauges; (a) measurement of acceleration and (b) measurement of dynamic strain.

modal strain patterns of the bridge. Subsequently, these updated modal properties were validated by comparing them with results from dynamic measurements, which will be detailed in a later section. Although the bridge is a prestressed concrete box girder, explicit modeling of prestress tendons and force effects was not included in this study. This simplification was adopted because the focus is on in-service dynamic behavior under ambient vibration, where the section remains uncracked and prestress effects primarily serve to maintain structural integrity rather than directly influence modal strain. Prestressing does slightly alter the effective stiffness of the section and contributes to keeping the neutral axis within the uncracked elastic range, but under low-amplitude vibration, its influence on NA position is relatively small and was not the dominant factor in this analysis.

4. SHM using dynamic measurements

To assess the dynamic characteristics of the targeted bridge, a SHM system, equipped with accelerometers and piezoresistive strain gauges for measuring acceleration and dynamic strain (see SHM in Fig. 1), was installed along the length of the box girder. The dynamic measurements of the bridge under ambient vibration and during regular operation were conducted to identify the frequencies, mode shapes, modal strain, and neutral axis depth. A frequency-domain technique was employed to extract these modal properties from the measurement data. The measurement results were then compared with both the as-designed and as-is models to validate and refine the baseline DT model of the bridge.

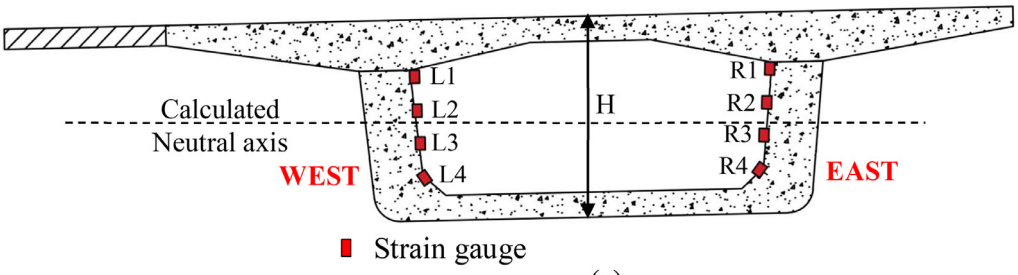
4.1. Measurement of acceleration under ambient vibration

To identify the natural frequencies and mode shapes of the bridge, acceleration-based measurements were conducted using two sets of uniaxial accelerometers in the vertical direction, as shown in Fig. 10(a). One set was positioned 88.7 m from the bridge's left side and designated as the reference station, indicated by a blue cross in Fig. 10(a). This location was chosen both for its ability to capture the 1st fundamental mode shapes of a three-span bridge and for its reduced sensitivity to local defects in the structure, thus ensuring consistent acceleration measurements. The second set of accelerometers was utilized at moving stations, consisting of 25 locations strategically placed with intervals of 7.5 m and 10 m in the side and middle spans, respectively. These moving stations are marked by red circles in Fig. 10(a). To record acceleration data, two data acquisition systems were set up at both the reference and moving stations. A voltage pulse signal generator was employed to ensure time synchronization among the data acquisition units [41].

The accelerometer at the reference station features an acceleration measurement capacity of ±0.5 g, whereas the accelerometers at the moving stations have a capacity of ±0.25 g. Each measurement set at the various moving stations lasted 5 min with a sampling rate of 100 Hz. Under the service response, the natural frequency of the bridge at each location was calculated using the Pwelch windowing function in MATLAB software [42]. Mode shapes were determined by normalizing the peak magnitudes of Fourier Response Functions between the moving and reference acceleration data.

4.2. Measurement of dynamic strains under ambient vibration

Segmental box girders in a balanced cantilever system are prone to mid-span cracking and deflection due to environmental factors, materials aging, or overloading, all of which can significantly impacting the bridge's ultimate flexural capacity [43–46]. Therefore, the neutral axis (NA), influenced by material and geometric properties and loading conditions, is critical in assessing the stiffness and flexural performance necessary for structural integrity evaluations [24,47]. To monitor the neutral axis, dynamic strain measurements in critical zones were conducted using a piezo-resistive strain gauge (see Fig. 1) with a resolution of 0.02 με and a gauge factor of approximately 120. This represents an enhancement 60 times greater than that of traditional metal foil gauges, as noted in previous studies [48,49]. The high resolution of these gauges enabled precise capture of sub-micron strains during operational vibrations. In total, 336 piezo-resistive strain gauges were installed at 42 locations, illustrated in Fig. 10(b). To systematically record the linearly distributed strain across each cross-section, eight strain gauges, la-



(a)



(b)

Fig. 11. Location of strain gauges inside the box girder; (a) arrangement of strain gauges at a cross-section, (b) installed strain gauges at the bridge.

beled R1-R4 on the right and L1-L4 on the left sides of the bridge, were arranged vertically as depicted in Fig. 11(a) with the installed strain gauges in Fig. 11(b). Under the operational condition, strain measurements were conducted at a sampling frequency of 50 Hz with each test lasting 30 min. Then, cross power spectral density (CPSD) analyses were performed between the acceleration and strain data, and results were normalized against the reference CPSD data. The normalized Fourier strain profiles for each location were then derived by averaging the outcomes. Utilizing these averaged results, the modal strain patterns of the bridge in various modes were constructed, incorporating the phase shifts observed between the acceleration and strain data. While dynamic strain responses are theoretically sensitive to localized damage, their practical interpretation is often hindered by variations in unknown or uncontrolled excitation conditions. The use of normalized modal strain (NMS) allows for consistent comparison of strain distribution patterns by eliminating the dependency on excitation magnitude. Although this normalization may reduce the direct sensitivity to damage magnitude, it enhances effectiveness and enables correlation with numerical predictions, making it a valuable tool for structural assessment under operational loads [45].

4.3. Identification of normalized modal strain

To ensure proper synchronization between acceleration and strain data—acquired from two separate data loggers—a common voltage pulse was generated and transmitted to both systems. This pulse functioned as a time stamp, enabling precise temporal alignment of the datasets for the CPSD analysis. With the synchronized datasets established, the subsequent step involved extracting the modal strain characteristics of the bridge through frequency-domain analysis. The modal strain of the bridge was constructed based on the strain frequency response functions by using the CPSD technique [50,51] between the data in time-domain (t) of reference signal $x(t)$ and rover signals $y(t)$. In this study, the acceleration response was selected as the reference signal due to its lower sensitivity to local and mild structural damage compared to the strain data. The strain data measured at different locations along the bridge, as shown in Fig. 10(b), were considered as rover signals.

To maintain compatibility in the CPSD analysis, the acceleration data—originally sampled at a higher rate—was down-sampled to match the sampling frequency of the strain measurements. This adjustment preserved the frequency-domain integrity of the signals and enabled accurate computation of CPSD functions. To mitigate variance errors, the CPSD results from all measurement points were averaged across segments.

The cross-correlation function of reference station data $x(t)$ and rover station data $y(t)$ is defined by:

$$R_{xy_r} = \frac{1}{N}\sum_{s=1}^{N-1} x_s y_{s+r} \tag{1}$$

where $r = 0, 1, 2,\ldots,(N\text{-}1)$; N is a total of number segments. Assuming rover signal $y(t)$ is a periodic function according to the Fourier series assumption defined by:

$$y_{s+r} = y_{s+r-N} \tag{2}$$

where $s + r \geq N$; Then the CPSD (S_{xy_k}) and (S_{xy_x}) at k^{th} frequency are presented by:

$$S_{xy_k} = \frac{1}{N}\sum_{s=1}^{N-1} R_{xy_r} e^{\frac{-i2\pi kr}{N}} = \bar{X}_k Y_k \tag{3}$$

$$S_{xx_k} = \frac{1}{N}\sum_{s=1}^{N-1} R_{xx_r} e^{\frac{-i2\pi kr}{N}} = \bar{X}_k X_k \tag{4}$$

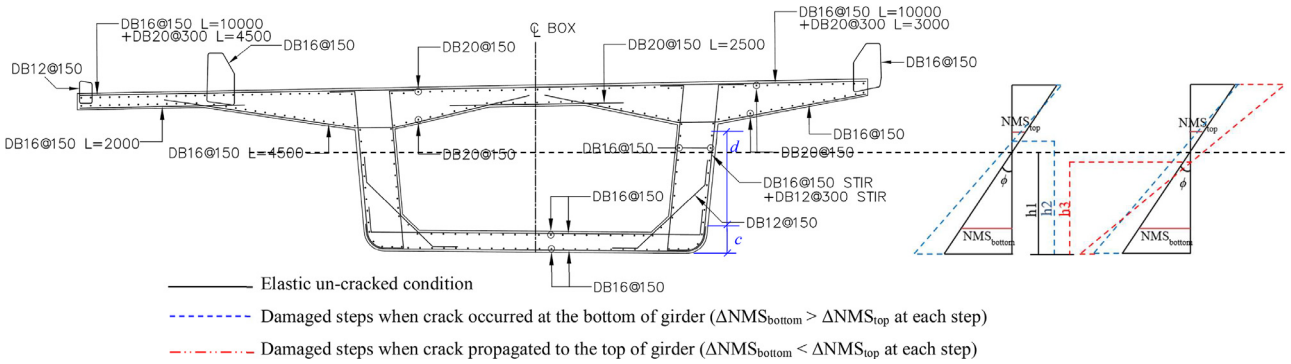


Fig. 12. Shifting of the neutral axis of the girder's cross-section.

where $k = 0, 1, 2,\ldots,(N\text{-}1)$ and $\bar{X}_k$ is the complex conjugate of X_k; X_k and Y_k are complex form of Fourier transform of $x(t)$ and $y(t)$ respectively at k^{th} frequency.

The Fourier magnitude of averaged CPSD of ith signal of jth mode was normalized with the Fourier magnitude of the CPSD of the reference signal itself to construct the normalized modal strain (NMS) following equation:

$$NMS_{ij} = \frac{\left|S_{xyk_j}\right|_i}{\left|S_{xxk_j}\right|} = \frac{\left|Y_{k_j}\right|_i}{\left|X_{k_j}\right|} \tag{5}$$

To achieve data consistency during the NMS computation, it is important to note that the top and bottom strain signals used to determine the position of the neutral axis were collected simultaneously using the same data acquisition system. Because these signals were recorded on a synchronized hardware platform, desynchronization between them is not expected, thus preserving the accuracy of the strain phase relationships and magnitude comparisons.

Finally, the modal strain vector can be computed by multiplying the normalized Fourier magnitude of each ith signal by the cosine of its Fourier phase θ_{xy_i} as:

$$\varphi_i = NMS_{ij}\cos\left(\theta_{xy_i}\right) \tag{6}$$

4.4. Identification of neutral axis depth obtained from NMS

In this study, the neutral axis (NA) was determined based on the NMS extracted at different frequency modes, allowing for the separation of the NA results due to bending deformation about the out-of-plane axis from those due to torsional deformation. The position of the NA in the cross-section is defined as the location where normal stress and strain are zero [24]. To calculate the NA of the section, it is essential to measure stress or strain values at a minimum of two points located on opposite sides of the NA. On the measurement, strain readings were taken at the upper and lower portions of the webs. These areas were selected due to their smooth surface, which is optimal for precise strain measurement at the designated section locations. Additionally, strain values were taken 250 mm from the top and bottom of the webs for these locations in both the as-designed and as-is models. Then, the NMS at the top (NMS_{top}) and bottom (NMS_{bottom}) of the cross-section as depicted in Fig. 12 can be used to calculate the NA depth from the bottom of the cross-section (h) as follows:

$$h = c + \frac{NMS_{bottom}}{NMS_{top} + NMS_{bottom}}\, d \tag{7}$$

where c is the distance from the bottom surface of cross-section to the location of NMS_{bottom}, d is the distance between NMS_{bottom} and NMS_{top} locations.

As the NA location correlated to the centroid of stiffness of a cross-section, the change in the NA position can indicate unusual behaviors in that section due to damage. Fig. 12 depicts the shifting of the neutral axis in a girder's cross-section under different conditions. In the elastic un-cracked condition shown by the solid black line (h_1), the neutral axis remains in its original, undamaged position. When cracks occur at the bottom of the girder, illustrated by the dashed blue lines (h_2), the neutral axis shifts upwards as the change in the neutral moment of strain at the bottom (ΔNMS_{bottom}) exceeds that at the top (ΔNMS_{top}). Conversely, as cracks propagate to the top of the girder, indicated by the dash-dot red lines (h_3), the neutral axis moves downwards because the strain alterations become more significant at the top (ΔNMS_{top}) than at the bottom (ΔNMS_{bottom}). Although the cross-section of a concrete box girder may experience temperature gradients that can alter stress distribution and affect the neutral axis position, temperature measurements were not incorporated in this study. To minimize such effects, all field data were collected during nighttime hours when thermal variations are minimal and relatively stable.

5. Comparison of bridge's dynamic behavior from DT-based models and dynamic analysis

In this section, the comparison of the bridge's dynamic properties between 3D-FEM and measurement results was conducted to update the model accordingly. Frequency domain analysis was performed in the 3D-FEM to identify the frequencies, mode shapes, and modal strain patterns.

5.1. Fundamental frequencies and mode shapes of the targeted bridge

Modal analyses were conducted using both the as-designed and as-is models. Since this study concentrated on flexural strain and the main observations involved vertical bending modes, only these specific modes were analyzed. Therefore, two vertical bending modes, namely 1st and 2nd bending modes, were identified using modal analysis under operational condition, and the comparison of the fundamental frequency results between the 3D-FEM and measurement results is presented in Table 2. With updated concrete properties and geometric details, the first two vertical bending frequencies of the as-is model closely matched the measured results. The natural frequency of the 1st bending mode in the as-is model, at 1.62 Hz, closely aligned with the 1.63 Hz measured directly using the frequency-domain technique. In contrast, the natural frequency of the 1st bending mode in the as-designed model was 1.59 Hz. By updating the concrete modulus of elasticity based on the UPV test and geometric details from the 3D-LiDAR scan, the frequencies of the 1st and 2nd bending modes in the as-is model more closely matched the measured frequencies. This alignment indicates successful validation of the updated model. Only the first two vertical bending modes were presented, as they exhibited the highest mass participation factors and best represented the bridge's global flexural behavior under ambient excitation. The torsional modes were excluded from the anal-

Table 2
Natural frequencies of the bridge obtained from 3D-FEM and dynamic measurement.

Fundamental modal frequencies	Natural frequency from 3D-FEM (Hz)		Natural frequency from measurement (Hz)
	As-designed drawing	As-is condition	
1st bending mode	1.59	1.62	1.63
2nd bending mode	2.32	2.41	2.46

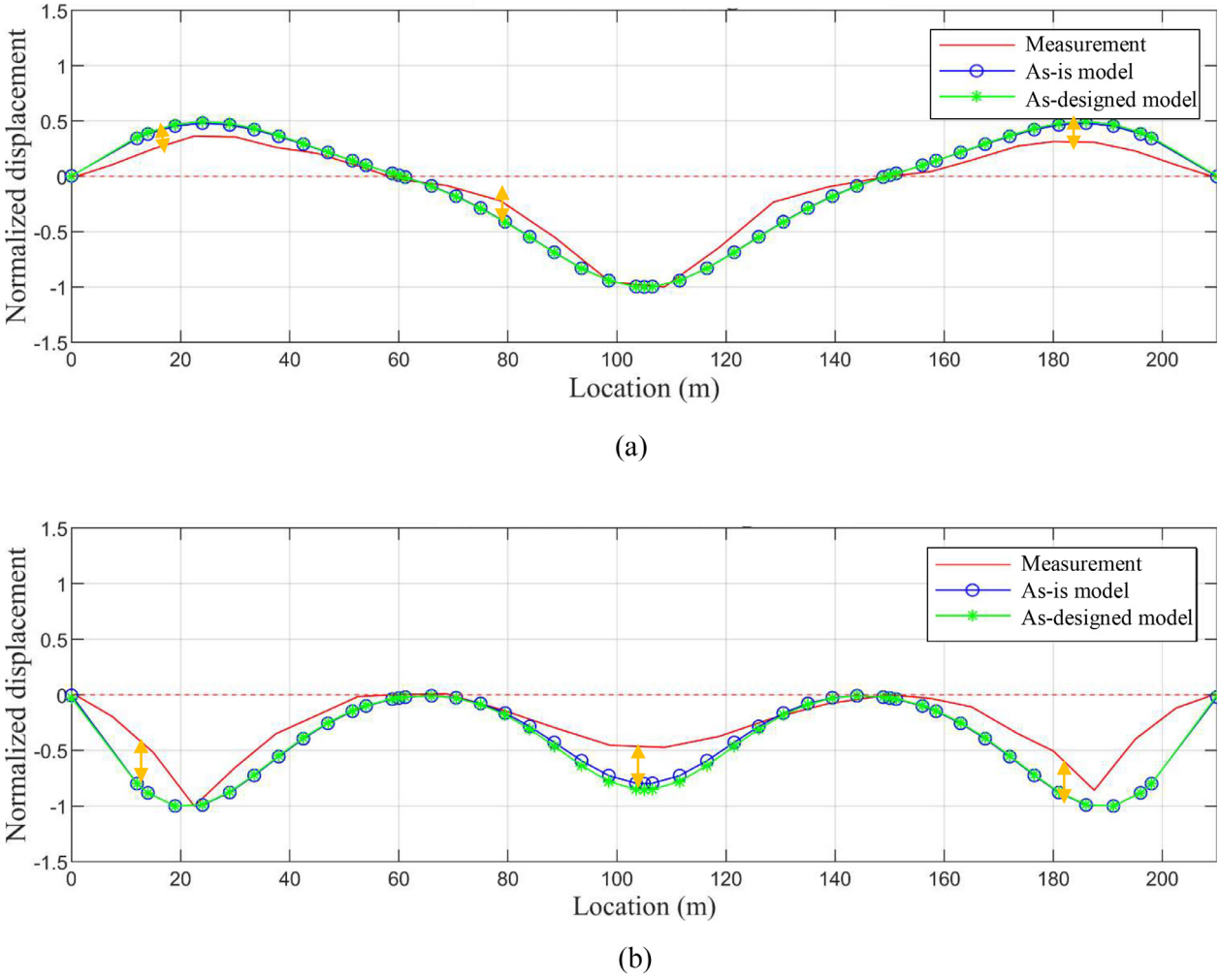


Fig. 13. Displacement mode shapes in the vertical direction; (a) 1st bending mode and (b) 2nd bending modes.

ysis because their contribution was negligible due to the bridge's high torsional stiffness, which is characteristic of box girder sections.

Compared to natural frequencies, mode shapes can be a more sensitive indicator of local structural damage, as they offer a detailed depiction of structural deformability at specific frequencies. Therefore, to accurately assess the extent of damage or irregularities in the bridge, the vertical bending mode shapes were extracted from the numerical models. These were then compared with the mode shapes derived from acceleration measurements to validate the model and identify any discrepancies. Fig. 13 presents the displacement mode shapes in the vertical direction of the bridge (1st and 2nd bending modes) from the as-designed model, the as-is model, and the measurement results. These mode shapes primarily illustrate the symmetric vertical bending vibrations that are significant at the bridge's midspan. As shown in Fig. 13, the mode shapes from the as-is model align well with the measured results, particularly at midspan where vertical displacement dominates. However, minor deviations at the side spans and segment joints highlight the limitations of mode shapes in detecting localized changes. Similar to natural frequencies, mode shapes are insensitive to minor damage and can be influenced by ambient conditions or boundary variability. While this comparison validates the as-is model for capturing global behavior, it also underscores the need for more sensitive indicators. To address this, the study incorporates normalized modal strain (NMS), derived from both acceleration and strain data, which offers improved sensitivity to localized stiffness variations and damage not captured by displacement mode shapes alone.

Initially, there were significant discrepancies in the fundamental frequencies and mode shapes between the as-designed model and the measurement results. However, after updating the geometry and concrete properties based on the 3D-LiDAR scan and UPV test, the as-is digital model showed excellent consistency with the measurement results. This indicates that the bridge's geometry and material properties have

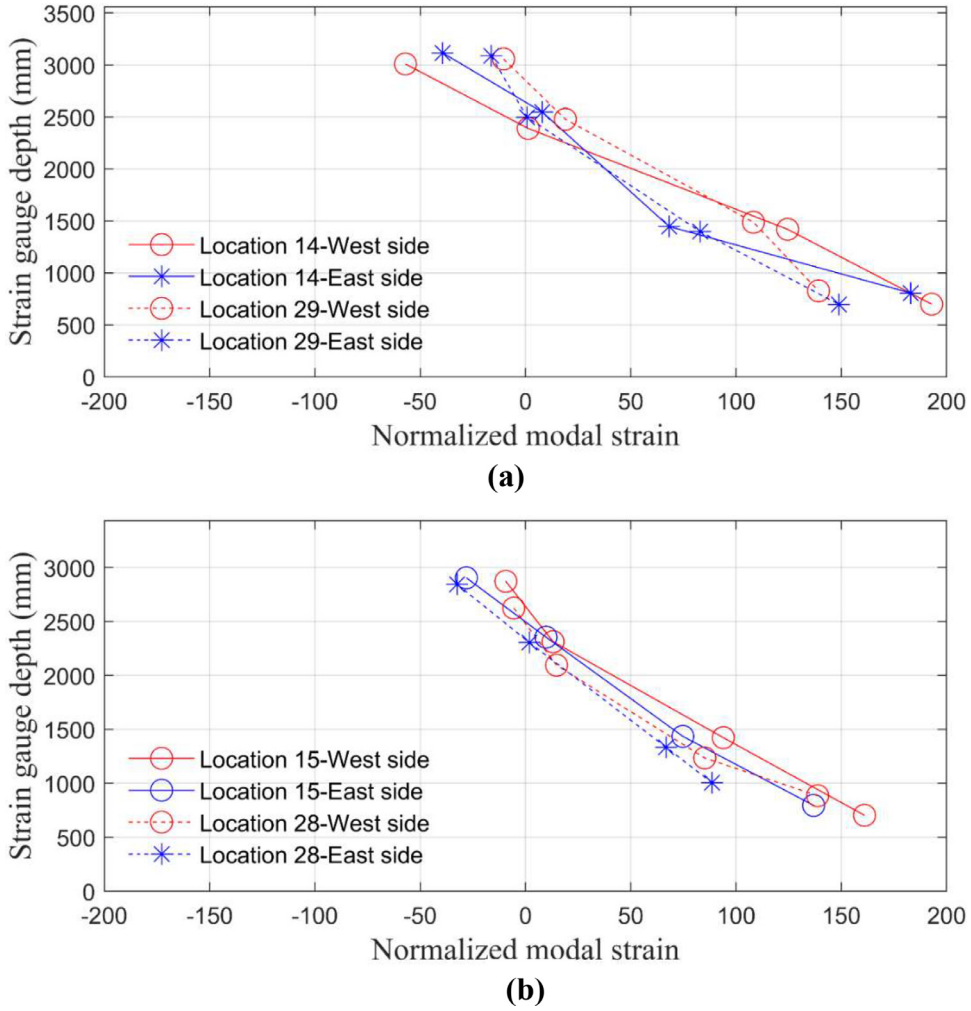


Fig. 14. Strain distribution at specified locations symmetrically along the bridge direction; (a) Location 14 and 29, and (b) Location15 and 28.

changed since its construction. Accurately capturing the bridge's current condition in the as-is digital model is crucial for precise evaluations and assessments. The as-is model now serves as a reliable baseline for further analysis.

5.2. Distribution of NMS

At each monitoring station (see Fig. 10(b)), four strain gauges were vertically aligned on each web, and at most of these stations, the NMS strain distribution demonstrated a linear relationship, as illustrated for locations 14, 15, 28, and 29 in Fig. 14. It should be noted that locations 14 and 29, as well as locations 15 and 28, are symmetrical along the direction of the bridge, as presented in Fig. 10(b). The magnitude and direction of NMS at locations 14 and 15 were found to be the same as at locations 29 and 28, respectively, as displayed in Fig. 14(a) and (b). Across the same monitoring station, the NMS distribution was consistent between the West and East webs, enabling the derivation of a linear strain relationship from the bottom to the top of each web. To quantify this relationship, the least squares method was applied to formulate the linear curvature equation of NMS distribution for each web location [52]. This approach minimizes the residuals between the data points and the regression line, offering an easy-to-interpret model that simplifies ongoing structural assessments and predictions. Minor deviations from linearity observed in some sections may be attributed to the variability of semi-conductive strain gauges, which are highly sensitive but prone to slight inconsistencies due to gauge factor variations. As such, the symmetry across the East and West sides at corresponding points suggests uniform structural dynamics and integrity, critical for validating the bridge's structural behavior under the operational load. Importantly, while the original dynamic strain data contains valuable damage-sensitive information, it is influenced by unknown or variable excitation levels in ambient conditions. NMS addresses this challenge by normalizing the frequency-domain strain data, emphasizing the relative shape and distribution of modal strain rather than absolute amplitudes. This allows for reliable comparison with numerical results and highlights deviations due to structural irregularities. However, this normalization inherently filters out amplitude-based differences that could indicate the severity of localized stiffness loss. Therefore, while NMS is effective for spatial pattern recognition and structural consistency assessment, it should be complemented by absolute strain analysis or calibrated damage indices in future work to improve sensitivity to minor or incipient damage.

5.3. Modal strain patterns

The modal strain patterns of the bridge were extracted from modal analysis in the as-designed and as-is models to evaluate the structural performance and identify discrepancies between the intended and actual behavior. Figs. 15 and 16 illustrate the modal strain patterns of the

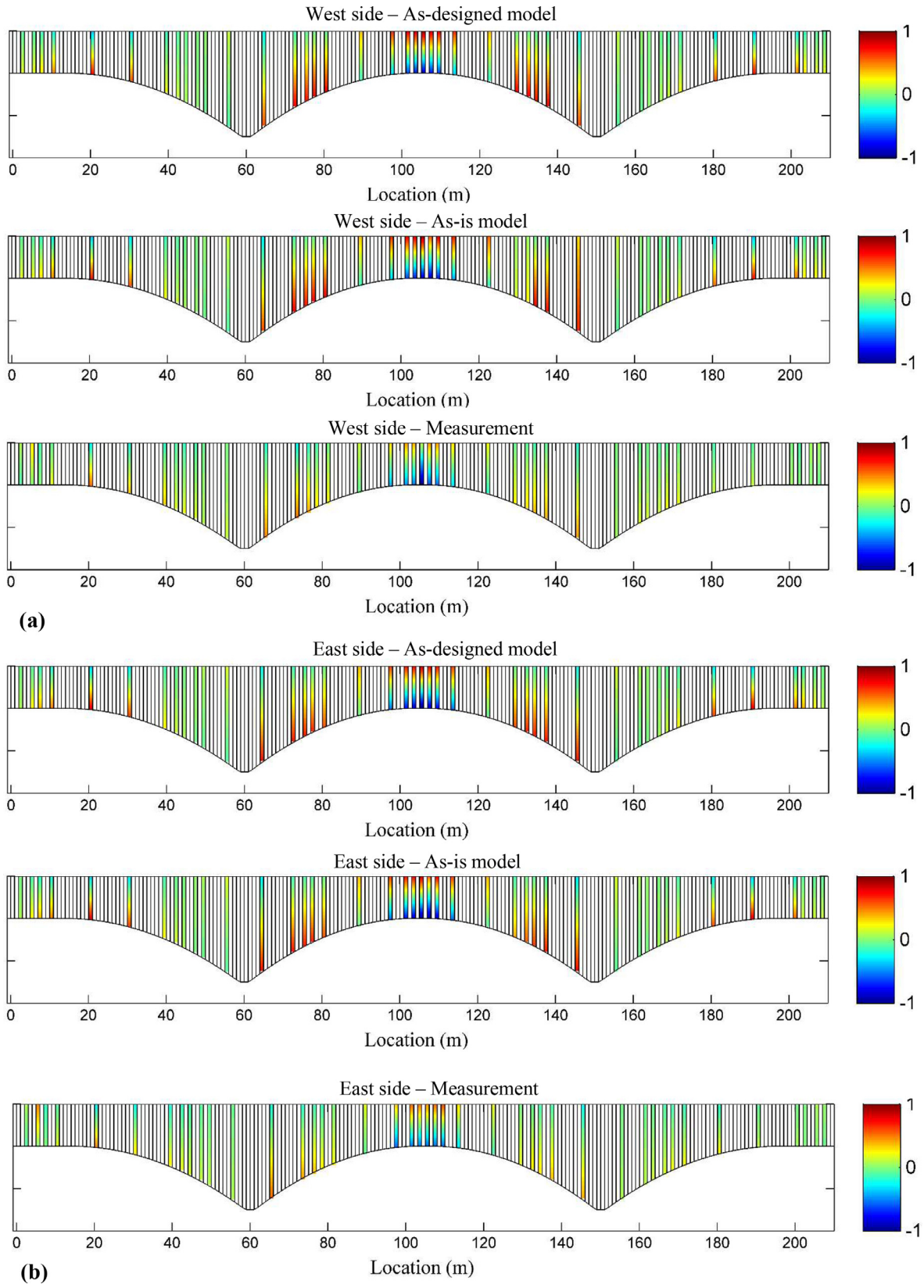


Fig. 15. Modal strain patterns for the 1st bending mode in the vertical direction obtained from numerical analysis and measurement; (a) West side and (b) East side of the bridge.

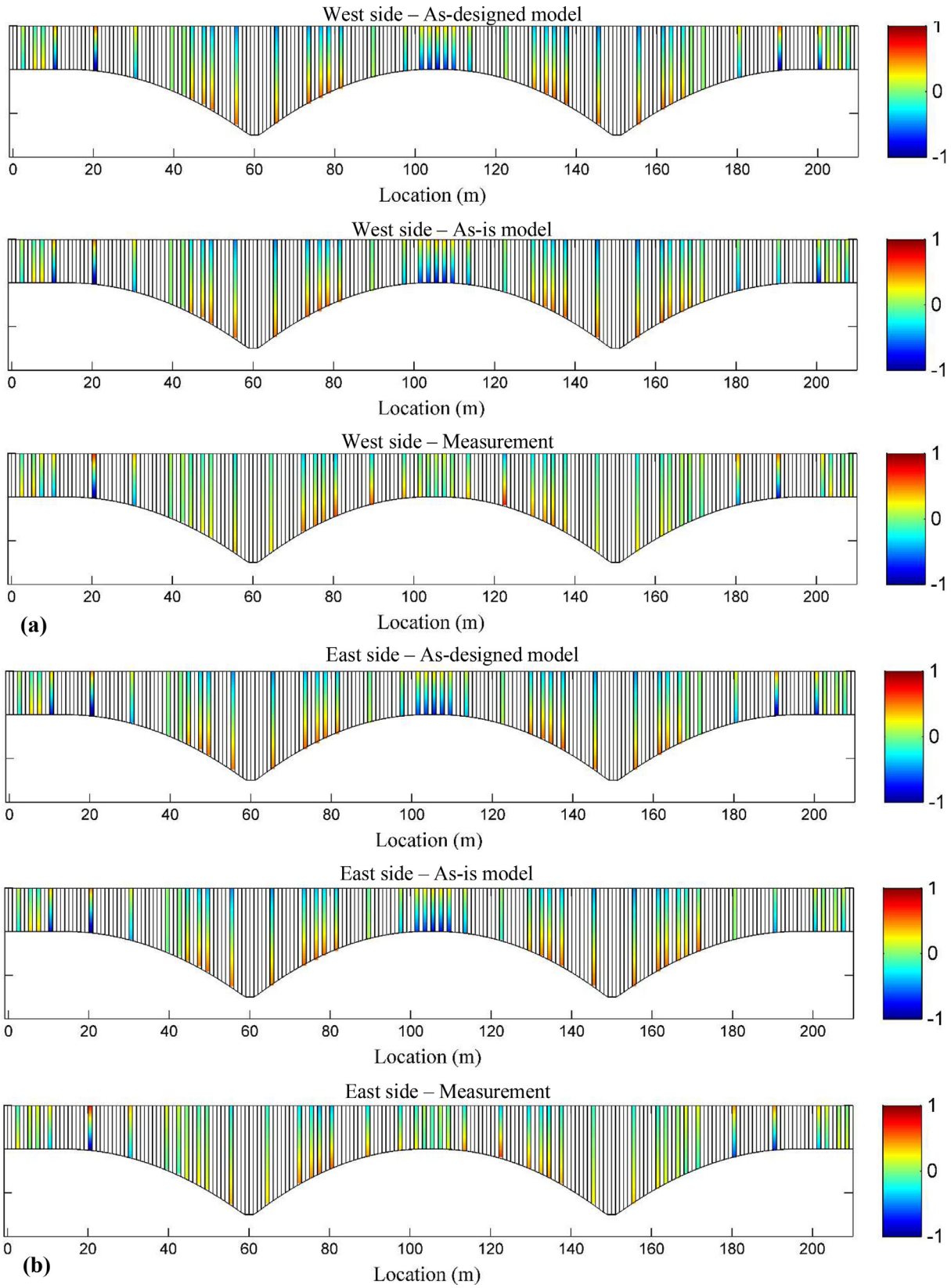


Fig. 16. Modal strain patterns for the 2nd bending mode in the vertical direction obtained from numerical analysis and measurement; (a) West side and (b) East side of the bridge.

bridge in the 1st and 2nd vertical bending modes under the operational vibration, respectively.

According to the comparison, the distributed trend of the modal strain pattern in the West and East web of the bridge from the field measurement showed good consistency with the numerical results. The symmetry of the bridge was evident in the strain distribution along its length during bending modes, consistent with the measured displacement mode shapes. The color representing the NMS magnitude in the measurements is lighter than that from the numerical model, except at the mid-span location. The lighter NMS color in the side spans may be attributed to potential minor cracks at the mid-span; however, verification was impeded by the mid-span's location over the river. It is also noted that this tendency for measured NMS magnitudes in the side spans to be lower than the numerical model corresponds with observations from the displacement mode shapes (Fig. 13), where measured displacements in similar regions also appeared smaller than model predictions. This suggests a consistent deviation where the model may be overestimating the response amplitude in these areas compared to the actual structure. If there is still a change in the NMS pattern, further study is required to check for damage in the middle span to verify the change in NMS.

The difference in NMS distribution may also be due to the imperfect support present in the model. Discrepancies of NMS obtained from measurement and FEM models at the span join location may arise due to the cantilevering construction method and the complexity of construction stages. These stages involve various time-dependent stressing effects, which can contribute to the observed differences. By calibrating the as-is model with measurement data, a more accurate predictive model can be developed for future heavy load scenarios and damage assessment.

As demonstrated in both measured and simulated patterns, NMS is a powerful indicator for assessing modal consistency and structural symmetry. Nevertheless, the potential limitation of reduced sensitivity to subtle damage severity due to amplitude normalization warrants attention. A more comprehensive evaluation approach could involve combining NMS with raw strain amplitude trends or regression-based neutral axis tracking to enhance diagnostic accuracy.

5.4. Shifting of NA depth

The NA depth serves as a critical parameter for evaluating the flexural behavior of the bridge and assessing potential structural changes over time. The NA depth was determined from NMS measurements at the top and bottom of the cross-sections for different bending modes. The results for the first two vertical bending modes (f = 1.635 Hz and f = 2.468 Hz) are presented in Fig. 17, illustrating comparisons between measured values and predictions from both the as-designed and as-is finite element models.

The measured NA depths exhibit a stable pattern along the bridge length, closely following the trends predicted by the numerical models. Variations in NA depth are observed at inflection points, where moment redistribution occurs, as expected. The as-is model demonstrates stronger agreement with the measured data compared to the as-designed model, particularly at midspans and inflection zones. This improved correlation suggests that updating the finite element model with geometric and material properties derived from 3D-LiDAR scans and UPV tests enhances its predictive accuracy.

It is also important to note that the NA positions differ between the 1st and 2nd bending modes due to the fundamental differences in their modal shapes and deformation patterns. The 1st bending mode, characterized by a single curvature over the span, leads to more uniform strain distributions and relatively consistent NA positions. In contrast, the 2nd bending mode introduces multiple inflection points, resulting in more localized curvature changes and dynamic strain concentration in different regions of the girder. These higher-mode effects tend to amplify differences in NA depth, especially in structurally complex or less stiffened sections. Thus, variations in NA depth across modes are not anomalies but rather theoretical outcomes of differing modal behavior.

Additionally, the observed fluctuations in NA depth align with strain distribution trends discussed in Section 5.3, reinforcing the consistency of the measured response. Importantly, no significant shifts in the NA position are detected, indicating that the overall flexural behavior of the bridge remains consistent with design expectations. The absence of abrupt deviations suggests that no major structural damage, such as excessive cracking or material degradation, has compromised the bridge's stiffness or load-carrying capacity.

The agreement between the measured NA depths and the as-is model validates the digital twin framework for this structure. By integrating real-time monitoring data with numerical modeling, the updated model provides a reliable baseline for ongoing SHM and facilitates early detection of potential structural changes. However, for future structural health monitoring and damage identification, the measured NA positions should serve as the primary benchmark. Unlike numerical predictions, field measurements capture the actual structural behavior under real operational and environmental conditions, including the effects of damage, construction variability, and material aging. While the digital twin model is invaluable for comparative analysis and forecasting, the real-time measurements provide the most accurate baseline against which changes and anomalies can be detected over time. Continuous monitoring of NA variations over time will be essential for tracking long-term performance and identifying emerging structural concerns, particularly in regions susceptible to environmental effects or high traffic loads.

6. Discussion

Accurately capturing the current state of the balanced cantilever bridge was critical for developing the DT-based SHM system in this study. 3D-LiDAR scanning effectively provided the as-is geometry, while UPV tests were essential for determining the in-situ concrete modulus of elasticity. The resulting detailed 3D point cloud model of the bridge's box girder demonstrated considerable advantages over conventional visual inspections regarding time efficiency, accessibility, and the level of detail achieved, affirming the value of these non-destructive techniques for creating robust "as-is" DT models.

A significant finding was the observed discrepancy between the measured as-is geometry and concrete modulus of elasticity compared to the original as-designed details. This highlights the imperative to develop DT models that reflect real-time structural behavior, necessitating updates after construction and throughout the service life of civil infrastructure to ensure the accuracy of structural assessments.

Validation of the "as-is" analytical model against field measurements was a cornerstone of this research. Modal properties extracted from vibration measurements, specifically natural frequencies and bending mode shapes, showed good agreement with the "as-is" model results. This synergy between as-is geometric and material data, integrated with experimental results, is crucial for validating numerical FE models intended for reliable bridge assessment.

Dynamic modal strain patterns and NA depth, derived by normalizing dynamic strain to acceleration data, emerged as sensitive indicators of structural behavior and correlated well with "as-is" model predictions. This confirms their utility for establishing a comprehensive baseline for future bridge inspections. However, some observable differences in NMS and NA magnitude persisted between experimental and numerical results, particularly at span join locations. These discrepancies may arise from uncaptured material defects or the complexities of modeling time-dependent effects from various construction stages—challenges inherent in numerical simulations. Such localized variations identify areas for further refinement in DT modeling, especially in the detailed representation of joints and material non-uniformities.

The enhanced diagnostic capabilities afforded by this validated "as-is" DT model, coupled with the sensitivity of parameters like NMS and

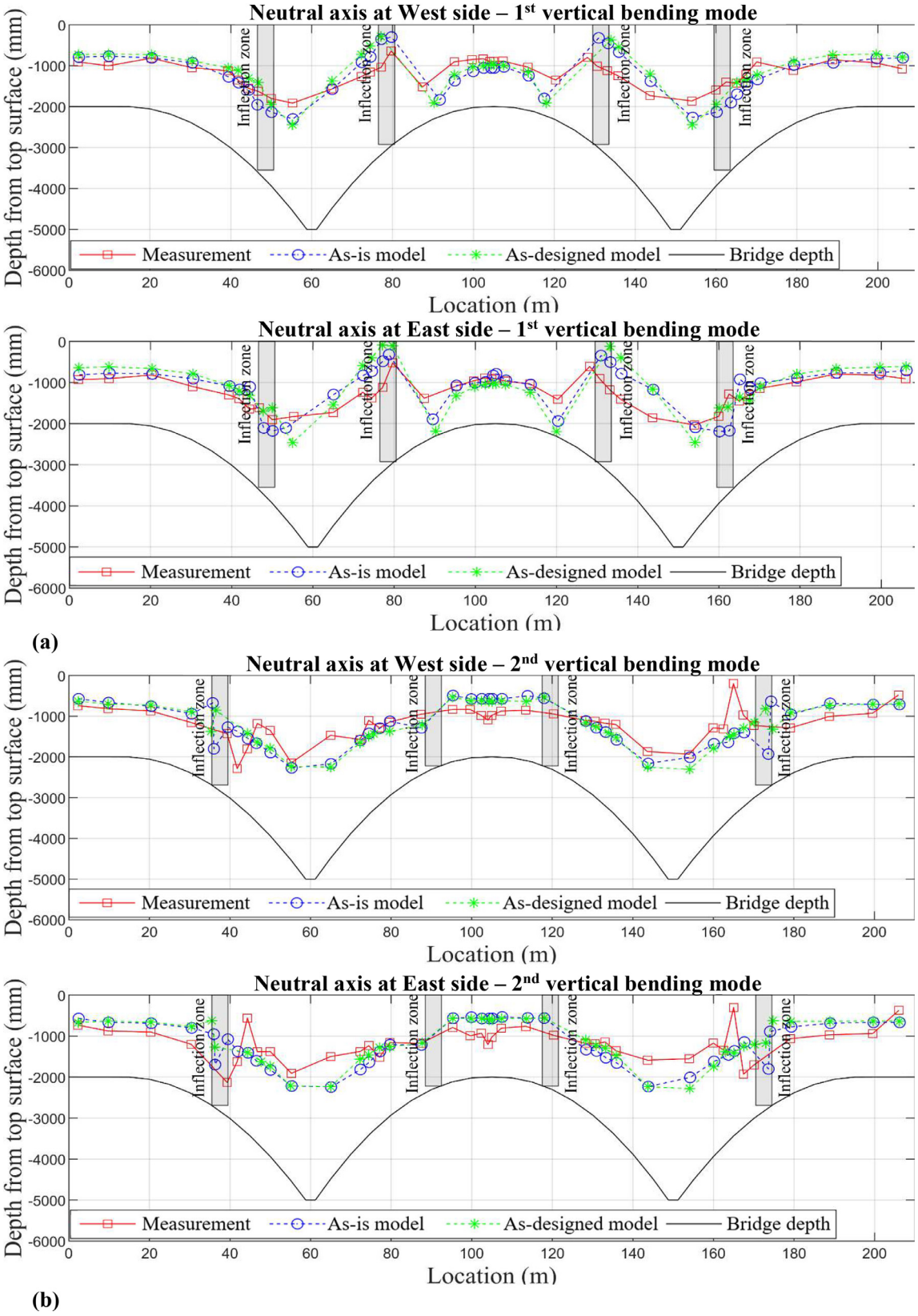


Fig. 17. NA depth due to the vibration in the vertical direction; (a) 1st bending mode (f = 1.635 Hz) and (b) 2nd bending mode (f = 2.468 Hz).

NA depth to local changes, directly contribute to the bridge's resilience. Early and accurate identification of baseline deviations, enabling proactive interventions, is crucial for minimizing damage escalation, repair costs, and downtime, thereby protecting critical infrastructure from cascading effects [53] and extending its functional service life. This proactive stance is a cornerstone of resilient infrastructure management, ensuring critical assets can better withstand and adapt to various performance challenges over time.

These refinements are particularly pertinent for segmented structures like the balanced cantilever bridge investigated. The inherent complexities of its construction, including segmental assembly and potential variations in cross-sectional geometry, strongly underscore the value of the "as-is" DT approach. Indeed, the improved correlation of the "as-is" model with measured natural frequencies and mode shapes, when compared to the as-designed model, clearly demonstrates DT's superior capacity to capture these unique and critical structural characteristics accurately.

While this study demonstrates the potential of the DT-SHM approach, it is important to acknowledge its current limitations to contextualize the findings. The 'as-is' model relies on key assumptions, such as not explicitly considering the time-dependent effects of prestress variation, which could influence long-term behavior. Furthermore, certain interpretations, such as linking NMS discrepancies to potential unverified cracks, remain speculative without direct physical confirmation. These aspects underscore that while the current DT provides a robust baseline, it represents a foundational stage.

Therefore, future research should aim to generalize these findings from this single case study by applying the SHM methodology to a broader range of bridge configurations. Long-term SHM is essential to verify the reliability of NA depth and NMS as indicators of local damage progression. Efforts should also be directed towards developing robust calibration and automated updating procedures for "as-is" models using continuous measurement results. Recognizing that advanced DT paradigms often feature more automated algorithms [54,55], future advancements could incorporate machine learning or Bayesian techniques to address the current limitations and enhance the dynamism of the Digital Twin [56–58].

7. Conclusions

This study successfully developed and applied a DT based SHM system for a balanced cantilever bridge. The primary conclusions are:

1. 3D-LiDAR scanning and UPV tests are effective methods for capturing the as-is geometry and concrete modulus of elasticity, respectively, facilitating the development of an accurate DT model that reflects the structure's current condition. The detail and efficiency of 3D-LiDAR offer considerable advantages over traditional inspection techniques.
2. Significant observed differences between the as-designed specifications and the as-is properties of the bridge, highlighting the necessity of creating and updating DT models throughout a structure's service life for reliable real-time behavioral assessment.
3. Field vibration measurements confirmed that the "as-is" DT model, by incorporating actual geometry and material properties, accurately predicted the bridge's key dynamic characteristics, including natural frequencies and bending mode shapes, thereby validating the numerical FE model.
4. Dynamic modal strain patterns and NA depth, derived from normalized field measurements, showed good agreement with the "as-is" model and serve as sensitive indicators, establishing a robust baseline for future SHM and damage detection efforts.
5. While the integrated DT and field measurement approach provides a valuable baseline for structural assessment, identified localized discrepancies highlight the need for further model refinement, particularly at complex joints and in material characterization.
6. Specifically for the balanced cantilever bridge investigated, the Digital Twin approach, when updated with as-is geometric and material data, effectively captured its unique dynamic characteristics and proved more accurate than the original design model. This underscores the critical importance of as-is modeling for complex structures like balanced cantilever bridges, where construction nuances and service-life changes can significantly alter structural behavior from initial design assumptions.

This research demonstrates the effectiveness of combining advanced measurement techniques with DT modeling for enhanced structural assessment and forms a foundation for future investigations into long-term monitoring and predictive maintenance of bridge infrastructure. Subsequent work can focus on integrating data-driven algorithms to further enhance the dynamism and predictive capabilities of this DT approach, moving towards more automated and optimized infrastructure management.

Relevance to resilience

This study advances infrastructure resilience through the integration of DT technology with SHM for a balanced cantilever bridge. This synergy facilitates real-time condition assessment and predictive maintenance, enabling proactive structural management. By leveraging 3D-LiDAR scanning, UPV tests, and dynamic strain measurements, the developed DT model accurately mirrors the bridge's as-is conditions. This accurate representation is fundamental for the early detection of potential structural vulnerabilities and for enabling timely, informed interventions.

The validated DT-SHM framework contributes significantly to resilience-based management by enhancing preparedness and facilitating a more effective response to potential structural issues. It aims to minimize downtime and support rapid functional recovery following disruptive events, extend the bridge's effective service life, and offers a scalable approach applicable to other critical urban infrastructure. Consequently, this research aids in maintaining the operational continuity of vital transportation assets, thereby reinforcing the overall resilience of urban networks.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Tidarut Jirawattanasomkul: Writing – original draft, Visualization, Validation, Resources, Project administration, Funding acquisition, Conceptualization. **Le Hang:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Supasit Srivaranun:** Writing – review & editing, Validation, Project administration, Conceptualization. **Suched Likitlersuang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Pitcha Jongvivatsakul:** Writing – review & editing, Validation. **Wanchai Yodsudjai:** Funding acquisition, Conceptualization. **Punchet Thammarak:** Software, Project administration, Methodology, Investigation, Data curation, Conceptualization.

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